

新东方烹饪学校大专化学B

GC卤大鹅班期末题

Part A: Multiple Choice – Select the one that is best in each case. (50 Pts)
Number of questions: 25

Points per question: 2

A1) Metals have different reactivity. Of the reactions below, only _____ is not spontaneous.

- A) $\text{Mg (s)} + 2\text{HCl (aq)} \rightarrow \text{MgCl}_2 \text{ (aq)} + \text{H}_2 \text{ (g)}$
- B) $2\text{Al (s)} + 6\text{HBr (aq)} \rightarrow 2\text{AlBr}_3 \text{ (aq)} + 3\text{H}_2 \text{ (g)}$
- C) $\text{Ni (s)} + \text{H}_2\text{SO}_4 \text{ (aq)} \rightarrow \text{NiSO}_4 \text{ (aq)} + \text{H}_2 \text{ (g)}$
- D) $2\text{Ag (s)} + 2\text{HNO}_3 \text{ (aq)} \rightarrow 2\text{AgNO}_3 \text{ (aq)} + \text{H}_2 \text{ (g)}$
- E) $\text{Zn (s)} + 2\text{HI (aq)} \rightarrow \text{ZnI}_2 \text{ (aq)} + \text{H}_2 \text{ (g)}$

A2) Which of the following are weak electrolytes?

HNO_3 HF NH_3 LiBr

- A) HNO_3 , LiBr
- B) HNO_3 , HF , NH_3 , LiBr
- C) HF , NH_3
- D) HF , LiBr
- E) HNO_3 , NH_3 , LiBr

A3) The first law of thermodynamics can be given as _____.

- A) $\Delta E = q + w$
- B) $\Delta H_{\text{rxn}}^\circ = \sum n\Delta H_f^\circ \text{ (products)} - \sum m\Delta H_f^\circ \text{ (reactants)}$
- C) for any spontaneous process, the entropy of the universe increases
- D) the entropy of a pure crystalline substance at absolute zero is zero
- E) $\Delta S = q_{\text{rev}}/T$ at constant temperature

A4) If matter is uniform throughout and cannot be separated into other substances by physical means, it is _____.

- A) a compound
- B) an element
- C) a heterogeneous mixture
- D) a homogeneous mixture
- E) either an element or a compound

A5) In general, as you go across a period in the periodic table from left to right:

- (1) the atomic radius _____;
 - (2) the electron affinity becomes _____ negative; and
 - (3) the first ionization energy _____.
- A) decreases, decreasingly, increases
 B) increases, increasingly, decreases
 C) increases, increasingly, increases
 D) decreases, increasingly, increases
 E) decreases, increasingly, decreases

A6) Viscosity is _____.

- A) the "skin" on a liquid surface caused by intermolecular attraction
 B) the resistance to flow
 C) the same as density
 D) inversely proportional to molar mass
 E) unaffected by temperature

A7) The element phosphorus exists in two forms in nature as white phosphorus and red phosphorus. These two forms are examples of _____.

- A) isotopes B) allotropes C) oxidation D) metalloids E) noble gases

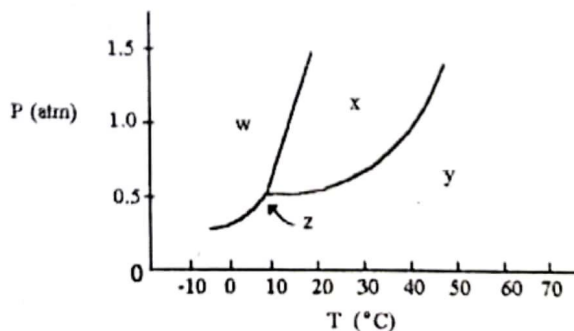
A8) In which orbital does an electron in a copper atom experience the greatest effective nuclear charge?

- A) $1s$
 B) $4s$
 C) $4p$
 D) $4d$
 E) $3d$

A9) A $0.200\text{ M K}_2\text{SO}_4$ solution is produced by _____.

- A) diluting 20.0 mL of $5.00\text{ M K}_2\text{SO}_4$ solution to 500.0 mL
 B) dissolving 20.2 g of K_2SO_4 in water and diluting to 250.0 mL , then diluting 25.0 mL of this solution to a total volume of 500.0 mL
 C) dilution of 1.00 mL of $250\text{ M K}_2\text{SO}_4$ to 1.00 L
 D) dilution of 250.0 mL of $1.00\text{ M K}_2\text{SO}_4$ to 1.00 L
 E) dissolving 43.6 g of K_2SO_4 in water and diluting to a total volume of 250.0 mL

A10) The normal boiling point of the substance with the phase diagram shown below is _____ $^{\circ}\text{C}$.



- A) 10 B) 15 C) 30 D) 40 E) 50

A11) When a system _____, ΔE is always negative.

- A) absorbs heat and does work
- B) absorbs heat and has work done on it
- C) gives off heat and does work
- D) gives off heat and has work done on it
- E) None of the above is always negative.

A12) Which formula/name pair is incorrect?

- A) HClO hypochlorous acid
- B) NaHCO₃ sodium bicarbonate
- C) N₂O₅ dinitrogen pentoxide
- D) Ni(CN)₂ nickel(II) cyanide
- E) Cu(ClO₄)₂ cupric perchlorite

A13) Of the following, only _____ is an extensive property.

- A) density
- B) volume
- C) boiling point
- D) freezing point
- E) temperature

A14) One million argon atoms is _____ mol (rounded to two significant figures) of argon atoms.

- A) 3.0
- B) 1.7×10^{-18}
- C) 6.0×10^{23}
- D) 1.0×10^{-6}
- E) 1.0×10^6

A15) Which statement below is incorrect?

- A) Of the three types of radioactivity characterized by Rutherford only gamma ray shows no response to an electric field.
- B) Atoms are composed of protons, neutrons, and electrons in Dalton's atomic theory.
- C) The gold foil experiment performed in Rutherford's lab led to the discovery of the atomic nucleus.
- D) The charge on an electron was determined in the Millikan oil drop experiment.
- E) Cathode rays travel in straight lines in the absence of electric or magnetic fields.

A16) What is the pH of a buffer solution that is 0.266 M in lactic acid and 0.111 M in sodium lactate? The K_a of lactic acid is 1.4×10^{-4} .

- A) 14.38
- B) 10.53
- C) 5.38
- D) 3.47
- E) 4.23

A17) The enthalpy change for converting 10.0 g of ice at -50.0°C to water at 50.0°C is _____ kJ. The specific heats of ice, water, and steam are 2.09 J/g-K, 4.18 J/g-K, and 1.84 J/g-K, respectively. For H₂O, $\Delta H_{\text{fus}} = 6.01 \text{ kJ/mol}$, and $\Delta H_{\text{vap}} = 40.67 \text{ kJ/mol}$.

- A) 12.28
- B) 4.38
- C) 3138
- D) 6.47
- E) 9.15

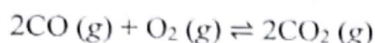
A18) In the Arrhenius equation,

$$k = Ae^{-E_a/RT}$$

_____ is the frequency factor.

- A) k
- B) A
- C) e
- D) E_a
- E) R

A19) Consider the following reaction at equilibrium.



$$\Delta H^\circ = -514 \text{ kJ}$$

Le Châtelier's principle predicts that the equilibrium mole fraction of CO (g) can be maximized by carrying out the reaction _____.

- A) at high temperature and high pressure
- B) at high temperature and low pressure
- C) at low temperature and low pressure
- D) at low temperature and high pressure
- E) in the presence of solid carbon

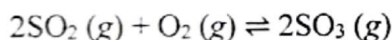
A20) Propane (C_3H_8) reacts with oxygen in the air to produce carbon dioxide and water. In a particular experiment, 38.0 grams of carbon dioxide are produced from the reaction of 22.05 grams of propane with excess oxygen. What is the % yield in this reaction?

- A) 38.0
- B) 57.6
- C) 66.0
- D) 86.4
- E) 94.5

A21) A 0.100 *m* solution of which one of the following solutes will have the highest vapor pressure?

- A) KClO_4
- B) $\text{Ca}(\text{ClO}_4)_2$
- C) $\text{Al}(\text{ClO}_4)_3$
- D) sucrose
- E) NaCl

A22) Consider the following equilibrium.



The equilibrium cannot be established when _____ is/are placed in a 1.0 -L container.

- A) 0.25 mol $\text{SO}_2 \text{ (g)}$ and 0.25 mol $\text{O}_2 \text{ (g)}$
- B) 0.75 mol $\text{SO}_2 \text{ (g)}$
- C) 0.25 mol of $\text{SO}_2 \text{ (g)}$ and 0.25 mol of $\text{SO}_3 \text{ (g)}$
- D) 0.50 mol $\text{O}_2 \text{ (g)}$ and 0.50 mol $\text{SO}_3 \text{ (g)}$
- E) 1.0 mol $\text{SO}_3 \text{ (g)}$

A23) The concentration of CO_2 in a soft drink bottled with a partial pressure of CO_2 of 6.5 atm over the liquid at 29 °C is $2.2 \times 10^{-1} \text{ M}$. The Henry's law constant for CO_2 at this temperature is _____.

- A) $2.2 \times 10^{-1} \text{ mol/L-atm}$
- B) $7.6 \times 10^{-3} \text{ mol/L-atm}$
- C) $5.6 \times 10^{-3} \text{ mol/L-atm}$
- D) $3.4 \times 10^{-2} \text{ mol/L-atm}$
- E) More information is needed to solve the problem.

A24) Which of the following choices has the compounds correctly arranged in order of increasing solubility in water? (least soluble to most soluble)

- A) $\text{CCl}_4 < \text{CHCl}_3 < \text{NaNO}_3$
- B) $\text{CH}_3\text{OH} < \text{CH}_4 < \text{LiF}$
- C) $\text{CH}_4 < \text{NaNO}_3 < \text{CHCl}_3$
- D) $\text{LiF} < \text{NaNO}_3 < \text{CHCl}_3$
- E) $\text{CH}_3\text{OH} < \text{CCl}_4 < \text{CHCl}_3$

A25) The $n = 2$ to $n = 6$ transition in the Bohr hydrogen atom corresponds to the _____ of a photon with a wavelength of _____ nm.

$$\Delta E = E_f - E_i = (-2.18 \times 10^{-18} \text{ J}) \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

- A) emission, 410
- B) absorption, 410
- C) absorption, 660
- D) emission, 94
- E) emission, 390

Part B: Short Answer – Write legibly and show all the work for all the steps in each problem. (20 Pts)

Number of questions: 5

Points per question: 4

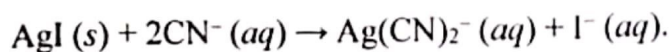
B1) Using the periodic table as a guide, write the condensed electron configuration and determine the number of unpaired electrons for the ground state of (a) Br, (b) Hf.

B2) Compare the radius and provide a brief explanation for each of the following pairs: (a) O^{2-} , O; (b) K^+ , Ca^{2+} .

B3) Select the stronger base in each of the following pairs: (a) BrO^- or ClO^- ; (b) BrO^- or BrO_2^- ; (c) HPO_4^{2-} or H_2PO_4^- ; (d) F^- or Cl^- .

B4) (a) Calculate the wavelength of a photon that has an energy of $5.25 \times 10^{-19} \text{ J}$. (b) What is the de Broglie wavelength (m) of a 2.0 -kg object moving at a speed of 50 m/s?

B5) (a) Use values of $K_{sp} = 8.3 \times 10^{-17}$ for AgI and $K_f = 1.0 \times 10^{21}$ for $\text{Ag}(\text{CN})_2^-$ to calculate the molar solubility of AgI in pure water. (b) Calculate the equilibrium constant for the reaction

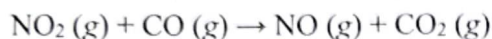


Part C: Long Answer-- Write legibly and show all the work for all the steps in each problem. Show both the initial equations used and the substituted values for mathematical relationships. Maintain correct units throughout your calculations and report your answers with the correct significant figures. (30 Pts)

Number of questions: 3

Points per question: 10

C1) The following data were measured for the reaction of nitrogen dioxide with carbon monoxide:



Experiment Number	[NO ₂] (M)	[CO] (M)	Initial Rate (M/s)
1	0.200	0.200	1.60×10^{-4}
2	0.200	0.400	1.60×10^{-4}
3	0.400	0.100	6.40×10^{-4}

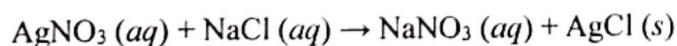
(a) Determine the rate law for this reaction.

(b) Calculate the rate constant magnitude with proper units.

(c) Calculate the half-life for NO₂ when [NO₂] = 0.400 M and [CO] = 0.400 M.

C2) The value of K_a for nitrous acid (HNO₂) at 25 °C is 4.5×10^{-4} . (a) Write the chemical equation for the equilibrium that corresponds to K_a . (b) By using the value of K_a , calculate ΔG° for the dissociation of nitrous acid in aqueous solution. (c) What is the value of ΔG at equilibrium? (d) What is the value of ΔG when $[\text{H}^+] = 5.0 \times 10^{-2} \text{ M}$, $[\text{NO}_2^-] = 6.0 \times 10^{-4} \text{ M}$, and $[\text{HNO}_2] = 0.20 \text{ M}$? Indicate if the forward reaction is spontaneous at 298 K.

C3) The precipitation reaction between AgNO₃ (aq) and NaCl (aq) proceeds as follows at 25°C:



(a) Write the net ionic equation of this reaction. (b) Given the data in the table below, calculate ΔH° for the net ionic equation of this reaction. (c) What would you expect for the value of ΔH° of the overall molecular equation compared to that for the net ionic equation? Explain. (d) Use the results from (b) and (c) along with data the table below to determine the value of ΔH_f° for AgNO₃ (aq).

Compound	AgNO ₃ (s)	Ag ⁺ (aq)	AgCl (s)	Cl ⁻ (aq)	NaCl (aq)	NaNO ₃ (aq)
ΔH_f° (kJ/mol)	-124.4	105.90	-127.0	-167.2	-407.1	-446.2

Fundamental Constants*

Atomic mass unit	1 amu = 1.660538921 × 10 ⁻²⁷ kg 1 g = 6.02214165 × 10 ²³ amu
Avogadro's number†	$N_A = 6.02214129 \times 10^{23}/\text{mol}$
Boltzman's Constant	$k = 1.3806488 \times 10^{-23} \text{ J/K}$
Electron charge	$e = 1.602176565 \times 10^{-19} \text{ C}$
Faraday's constant	$F = 9.64853365 \times 10^4 \text{ C/mol}$
Gas constant	$R = 0.0820582 \text{ L-atm/mol-K}$ $= 8.3144621 \text{ J/mol-K}$
Mass of electron	$m_e = 5.4857990946 \times 10^{-4} \text{ amu}$ $= 9.10938291 \times 10^{-31} \text{ kg}$
Mass of neutron	$m_n = 1.008664916 \text{ amu}$ $= 1.674927351 \times 10^{-27} \text{ kg}$
Mass of proton	$m_p = 1.007276466 \text{ amu}$ $= 1.672621777 \times 10^{-27} \text{ kg}$
Pi	$\pi = 3.1415927$
Planck's constant	$h = 6.62606957 \times 10^{-34} \text{ J-s}$
Speed of light in vacuum	$c = 2.99792458 \times 10^8 \text{ m/s}$

activity series of metals in aqueous solution

metal	oxidation Reaction
Lithium	$\text{Li (s)} \rightarrow \text{Li}^+ (\text{aq}) + e^-$
Potassium	$\text{K (s)} \rightarrow \text{K}^+ (\text{aq}) + e^-$
Barium	$\text{Ba (s)} \rightarrow \text{Ba}^{2+} (\text{aq}) + 2e^-$
Calcium	$\text{Ca (s)} \rightarrow \text{Ca}^{2+} (\text{aq}) + 2e^-$
Sodium	$\text{Na (s)} \rightarrow \text{Na}^+ (\text{aq}) + e^-$
Magnesium	$\text{Mg (s)} \rightarrow \text{Mg}^{2+} (\text{aq}) + 2e^-$
Aluminum	$\text{Al (s)} \rightarrow \text{Al}^{3+} (\text{aq}) + 3e^-$
Manganese	$\text{Mn (s)} \rightarrow \text{Mn}^{2+} (\text{aq}) + 2e^-$
Zinc	$\text{Zn (s)} \rightarrow \text{Zn}^{2+} (\text{aq}) + 2e^-$
Chromium	$\text{Cr (s)} \rightarrow \text{Cr}^{3+} (\text{aq}) + 3e^-$
Iron	$\text{Fe (s)} \rightarrow \text{Fe}^{2+} (\text{aq}) + 2e^-$
Cobalt	$\text{Co (s)} \rightarrow \text{Co}^{2+} (\text{aq}) + 2e^-$
Nickel	$\text{Ni (s)} \rightarrow \text{Ni}^{2+} (\text{aq}) + 2e^-$
Tin	$\text{Sn (s)} \rightarrow \text{Sn}^{2+} (\text{aq}) + 2e^-$
Lead	$\text{Pb (s)} \rightarrow \text{Pb}^{2+} (\text{aq}) + 2e^-$
Hydrogen	$\text{H}_2 (\text{g}) \rightarrow 2\text{H}^+ (\text{aq}) + 2e^-$
Copper	$\text{Cu (s)} \rightarrow \text{Cu}^{2+} (\text{aq}) + 2e^-$
Silver	$\text{Ag (s)} \rightarrow \text{Ag}^+ (\text{aq}) + e^-$
Mercury	$\text{Hg (l)} \rightarrow \text{Hg}^{2+} (\text{aq}) + 2e^-$
Platinum	$\text{Pt (s)} \rightarrow \text{Pt}^{2+} (\text{aq}) + 2e^-$
Gold	$\text{Au (s)} \rightarrow \text{Au}^{3+} (\text{aq}) + 3e^-$

Periodic Table of the Elements

Hydrogen 1 H 1.008																								Helium 2 He 4.0026																																																																																																																																																																																																											
Lithium 3 Li 6.94												Beryllium 4 Be 9.0122																								Boron 5 B 10.81												Carbon 6 C 12.011												Nitrogen 7 N 14.007												Oxygen 8 O 15.999												Fluorine 9 F 18.998												Neon 10 Ne 20.180																																																																																																																																			
Sodium 11 Na 22.990												Magnesium 12 Mg 24.305																								Aluminum 13 Al 26.982												Silicon 14 Si 28.085												Phosphorus 15 P 30.974												Sulfur 16 S 32.06												Chlorine 17 Cl 35.45												Argon 18 Ar 39.948																																																																																																																																			
Potassium 19 K 39.098												Calcium 20 Ca 40.078(4)												Scandium 21 Sc 44.956												Titanium 22 Ti 47.867												Vanadium 23 V 50.942												Chromium 24 Cr 51.996												Manganese 25 Mn 54.938												Iron 26 Fe 55.845(2)												Cobalt 27 Co 58.933												Nickel 28 Ni 58.693												Copper 29 Cu 63.546(3)												Zinc 30 Zn 65.38(2)												Gallium 31 Ga 69.723												Germanium 32 Ge 72.63												Arsenic 33 As 74.922												Selenium 34 Se 78.96(3)												Bromine 35 Br 79.904												Krypton 36 Kr 83.799(2)																							
Rubidium 37 Rb 85.468												Strontium 38 Sr 87.62												Yttrium 39 Y 88.906												Zirconium 40 Zr 91.224(2)												Niobium 41 Nb 92.906(2)												Molybdenum 42 Mo 95.96(2)												Technetium 43 Tc [97.91]												Ruthenium 44 Ru 101.07(2)												Rhodium 45 Rh 102.91												Palladium 46 Pd 106.42												Silver 47 Ag 107.87												Cadmium 48 Cd 112.41												Indium 49 In 114.82												Tin 50 Sn 118.71												Antimony 51 Sb 121.76												Tellurium 52 Te 127.60(3)												Iodine 53 I 126.90												Xenon 54 Xe 131.29																							
Cesium 55 Cs 132.91												Barium 56 Ba 137.33												57-70 *												Lutetium 71 Lu 174.97												Hafnium 72 Hf 178.49(2)												Tantalum 73 Ta 180.95												Tungsten 74 W 183.84												Rhenium 75 Re 186.21												Osmium 76 Os 190.23(2)												Iridium 77 Ir 192.22												Platinum 78 Pt 195.08												Gold 79 Au 196.97												Mercury 80 Hg 200.59												Thallium 81 Tl 204.38												Lead 82 Pb 207.2												Bismuth 83 Bi 208.98												Polonium 84 Po [209.98]												Astatine 85 At [209.99]												Radon 86 Rn [222.02]											
Francium 87 Fr [223.02]												Radium 88 Ra [226.03]												89-102 **												Lanthanum 103 La [202.11]												Rutherfordium 104 Rf [261.12]												Dubnium 105 Db [268.12]												Seaborgium 106 Sg [271.13]												Bohrium 107 Bh [270]												Hassium 108 Hs [277.15]												Mtnerium 109 Mt [276.15]												Darmstadtium 110 Ds [281.16]												Roentgenium 111 Rg [280.16]												Copernicium 112 Cn [285.17]												Ununtrium 113 Uut [284.18]												Flerovium 114 Fl [289.19]												Ununpentium 115 Uup [288.19]												Livermorium 116 Lv [293]												Ununseptium 117 Uus [294]												Ununoctium 118 Uuo [294]											

*lanthanoids

**actinoids

Lanthanum 57 La 138.91	Cerium 58 Ce 140.12	Praseodymium 59 Pr 140.91	Neodymium 60 Nd 144.24	Promethium 61 Pm [144.91]	Samarium 62 Sm 150.36(2)	Europium 63 Eu 151.96	Gadolinium 64 Gd 157.25(3)	Terbium 65 Tb 158.93	Dysprosium 66 Dy 162.50	Holmium 67 Ho 164.93	Erbium 68 Er 167.26	Thulium 69 Tm 168.93	Ytterbium 70 Yb 173.05
Actinium 89 Ac [227.03]	Thorium 90 Th 232.04	Protactinium 91 Pa 231.04	Uranium 92 U 238.03	Neptunium 93 Np [237.05]	Plutonium 94 Pu [244.08]	Americium 95 Am [243.06]	Curium 96 Cm [247.07]	Berkelium 97 Bk [247.07]	Californium 98 Cf [251.08]	Einsteinium 99 Es [252.08]	Fermium 100 Fm [257.10]	Mendelevium 101 Md [258.10]	Noberium 102 No [259.10]